

Kevin Paul Montealegre Melo

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Senior Mobile Software Engineer (Android & iOS)

Systems Engineer and Senior Mobile Developer with 6 years of experience architecting and maintaining large-scale applications within the fintech and corporate sectors. Expert in delivering secure, high-performance solutions for platforms with over 18 million active users. Specialist in Kotlin, Swift, and Jetpack Compose, with a deep focus on Clean Architecture (MVVM) and robust backend integration using Java Spring Boot. Proven ability to lead technical initiatives and optimise complex systems in high-pressure environments.

Core Technical Stack

- Mobile: Kotlin, Swift, Jetpack Compose, SwiftUI, Android SDK, Bluetooth BLE, iOS SDK, UIKit, Coroutines, Combine, Kotlin Multiplatform (KMP)
- Architecture & Logic: Clean Architecture, MVVM, SOLID, DDD, Modularisation, Design Patterns
- Backend & API: Java Spring Boot, RESTful APIs, Microservices, OAuth2, JWT, SOAP
- Testing & DevOps: JUnit, Espresso, MockK, Git, Docker, Jenkins, SonarQube, CI/CD Pipelines
- Databases: MySQL, PostgreSQL, Firebase, MongoDB, Room Database

Professional Experience

Dinastía, Medellín – Freelance

Dec 2025 – present

Senior Mobile Developer

- Architected and maintained the native iOS application using Swift, delivering core pet management features and ensuring a seamless user journey.
- Engineered high-performance API integrations and designed Kotlin-based backend services to support critical mobile functionalities.
- Optimised application performance through rigorous profiling and debugging, significantly reducing resource consumption and latency.
- Spearheaded collaboration with product and design teams to define technical requirements and deliver scalable, production-ready release

Grupo ASD, Bogota

Senior Android
Developer

May 2025 – Dec 2025

- Led the Android development for project SIBEL, implementing complex biometric solutions on industrial hardware (Aratek Marshall 8).
- Integrated the BMAPI SDK for fingerprint capture and verification, MRZ reading, and QR scanning, achieving 99.9% reliability in field conditions.
- Designed modern, responsive UIs using Jetpack Compose within an MVVM + Clean Architecture framework, improving system maintainability.
- Directed technical proofs of concept (PoC) and validated hardware-software compatibility for critical industrial environments.
- Implemented Bluetooth Low Energy (BLE) communication for device connectivity and handled secure data transfer between peripherals and the application.
- Developed and delivered a mobile solution for electoral scrutiny events for enterprise clients, supporting secure, on-site data collection and validation in high-stakes environments.

Valid, Bogota

Android Developer

Jun 2022 – Aug 2024

- Developed core functionalities for DaviPlata, a fintech application with over 18 million active users.
- Implemented critical features in Kotlin, strictly adhering to an MVVM architecture.
- Built modern user interfaces utilising Jetpack Compose.
- Optimised performance and resolved production incidents, directly impacting the stability of millions of daily transactions.
- Participated in security initiatives and the handling of sensitive data within a highly regulated fintech environment.
- Led the adaptation and delivery of the DaviPlata application for Huawei devices, ensuring full compatibility with HarmonyOS and environments without Google Play Services.

Qinaya, Bogota

Android Developer

Oct 2020 – Jan 2022

- Developed Android applications enabling remote desktop access.
- Implemented secure communication protocols and advanced authentication mechanisms.
- Optimised rendering performance across a wide range of Android devices.
- Applied data encryption to protect remote sessions and guarantee system security.
- Deployed and optimised the solution on devices powered by the S905X2 processor, contributing to the development of a launcher and Qinaya App used across thousands of devices.

Politecnico Grancolombiano, Bogota

Android Developer

May 2019 – Sep 2020

- Led the development of the institutional GoPoli application using Java and Kotlin.
- Implemented OAuth2 authentication and integrated internal campus services.
- Integrated push notifications using Firebase Cloud Messaging (FCM).
- Managed the full development lifecycle: design, development, testing, and deployment.
- Integrated hardware sensor APIs to track step counts, promoting physical activity across the university community

Education

BSc in Systems Engineering, 2023

Politécnico Grancolombiano, Bogotá, Colombia

Software Development Technician, 2017

SENA, Bogota, Colombia

Technical Skills

- **Android Development:** Kotlin, Java, Jetpack Compose, Android SDK, Android Studio, Material Design, Debugging, Testing (JUnit, Espresso, MockK), ExoPlayer, Android TV, Huawei HMS, Banking API Integration, Retrofit, Coroutines, Bluetooth, Kotlin Multiplatform (KMP), Industrial Hardware Optimisation
- **iOS Development:** Swift, iOS SDK, Xcode, UIKit, SwiftUI (Fundamentals), App Lifecycle, Navigation Patterns, REST API Integration, Local Persistence, MVVM for iOS, Apple Human Interface Guidelines, KMP Shared Module Integration
- **Architecture & Best Practices:** Clean Architecture, MVVM, SOLID Principles, Domain-Driven Design (DDD), Modularisation, Code Reviews, CI/CD, Dependency Management (Gradle, Swift Package Manager), Cross-Platform Architecture with KMP
- **Backend & Services:** Java Spring Boot, RESTful APIs, Microservices, Biometric SDK Integration (Aratek BMAPI, TrustFinger), SOAP/REST Services, Security & Authentication (OAuth2, JWT), Fault Tolerance
- **Databases:** MySQL, PostgreSQL, Firebase, MongoDB, Room Database, Data Modelling, Offline/Online Synchronisation
- **DevOps & Tools:** Git, Bitbucket, Jenkins, Docker, Gradle, Postman, SonarQube, Automated Testing, Log Monitoring, Firebase Crashlytics, Android Profiler
- **Web & Frontend:** Angular, Node.js, HTML, CSS, JavaScript, JSON/XML Processing, Responsive Design

Certifications

- Kotlin for Java Developers, JetBrains
- Kotlin Multiplatform Development
- Introduction to iOS Mobile Application Development
- Codingame Certification – Kotlin
- Swift for iOS Applications, Platzi
- MVVM Architecture Patterns for iOS and Android, Platzi
- Jetpack Compose for Android, Platzi
- Cybersecurity Fundamentals, Cisco
- Scrum Foundation Professional Certificate (SFPC)

